

User Flow Document

THE FARMER

The farmer is at the point of harvest. They open the Flutter app on their Android phone. Internet is not required at this point. They point the camera at their produce. The TFLite model runs directly on the phone and returns two things: a freshness score between 0 and 100 percent and an estimated number of days the produce will remain viable.

With that score, the farmer creates a listing. The farmer manually enters the crop type, quantity in kilograms, and the app automatically attaches their GPS location and the AI freshness score. The listing is sent to the backend which stores it in the database and generates a SHA-256 hash of that listing. A QR code is generated from that hash and attached to the listing.

The backend fetches the current regional price from the MOFA database and applies a freshness-weighted formula. It sets a price ceiling based on the MOFA price and freshness score. It also sets a soft floor at 60% of the MOFA reference price. The farmer can list their produce at any price between the soft floor and the ceiling. If they attempt to list below the soft floor, the system shows a warning that they are undervaluing their produce but allows them to proceed. This gives the farmer full agency while protecting them from irrational panic selling.

If the produce remains unsold and the freshness score is projected to drop below a threshold within 48 hours, the farmer receives an automated push notification warning them to act. When a buyer purchases the produce, the buyer's payment is held in the platform's internal account. The farmer does nothing and waits. When QR code confirmation happens at delivery or collection, the money is automatically released from the platform's internal account to the farmer.

THE BUYER

The buyer opens the marketplace on their Flutter app. They see produce listings in default chronological order. They can manually sort and filter by freshness score, crop type, region, and crop quantity. Every listing shows the AI-verified freshness score, the quantity the farmer listed, and the farmer's listed price all displayed together so the buyer evaluates quality, quantity, and price together. The buyer selects a listing and confirms purchase. Payment is made and held immediately in the platform's internal account. It does not go to the farmer yet. The system then asks the buyer one question: "Do you have your own transport to collect this produce?"

If the buyer answers YES, no driver is dispatched. The buyer arranges their own collection. The farmer is notified that the buyer will come directly. When the buyer arrives to collect, the farmer presents the QR code. The buyer scans it using the app. The app verifies that

the hash on the QR code matches the original listing hash in the database confirming the produce received matches what was listed. Confirmation is recorded and payment is released from the platform's internal account to the farmer.

If the buyer answers NO, the system identifies the nearest available registered driver whose truck capacity meets or exceeds the listed quantity and dispatches a notification to them. The buyer receives a push notification confirming that a driver has been contacted and is being arranged. When the driver accepts the job, the buyer receives another push notification confirming that their produce has been picked up and is in transit. When the driver delivers and QR code confirmation happens, payment is released to the farmer.

THE DRIVER

The driver is registered on the platform with their vehicle capacity in kilograms and their operating region. Drivers are verified through their registration status on the platform. When a buyer answers NO to the transport question, the backend identifies the nearest available registered driver whose truck capacity meets or exceeds the listing quantity. The backend sends them a job notification. Either via the Flutter app.

The notification contains:

- The farmer's GPS coordinates so the driver can navigate using their own maps application.
- The buyer's phone number so the driver can contact the buyer directly to agree on a cash delivery price before proceeding.
- The produce details including crop type and quantity in kilograms.
- The delivery destination.

The driver calls or messages the buyer to agree on a delivery price. Once agreed, the driver proceeds to the farmer's GPS location to collect the produce. The driver accepts or declines the job. If they decline, the system moves to the next nearest available driver automatically whose capacity also meets the requirement.

When the driver accepts, the buyer is notified immediately that their produce has been picked up and is in transit. The driver collects the produce from the farmer. At delivery, the buyer scans the QR code to verify the produce. The app verifies the hash matches the original listing. Delivery is confirmed. Payment is released from the platform's internal account to the farmer. The driver is paid directly in cash by the buyer upon delivery at the price they agreed beforehand. The platform does not handle driver payment in this version.

THE ADMINISTRATOR

The administrator works from a web-based dashboard on a desktop browser. They do not use the mobile app. When a new farmer, buyer, or driver registers via OTP phone verification, the admin reviews and approves the account before that user gains any platform access. No unverified user can transact. Driver registration includes their truck capacity in kilograms and operating region which the admin verifies.

The admin monitors all active listings, transactions, registered users, and flagged disputes across all regions. They can view the full audit trail every event from scan to delivery is SHA-256 hashed and timestamped and the chain of custody for any transaction can be verified at any time.

Every week the admin manually updates the regional crop price reference database using the weekly price bulletins published by MOFA Ghana. This keeps the price ceiling and soft floor calculations accurate throughout the season. The admin can resolve disputes between farmers, buyers, and drivers through the transaction dashboard.